Today: Regularity of Solutions to Heat Equation

1. Qualitative Regularity
2. Quantitative Regularity (regularity estimates)

<u>Theorem</u> If u solves

$$u_t - \Delta u = 0 \quad \text{in} \quad \mathbb{R}^n \times [0,T],$$

and $u(x,0)$ is bounded, then $u \in C^\infty(\mathbb{R}^n \times (0,T))$.

<u>Proof</u>

Let $g(x) = u(x,0)$.

Solution Formula:

$$u(x,t) = \int_{\mathbb{R}^n} \frac{e^{-|x-y|^2/4t}}{(4\pi t)^{n/2}} g(y)\,dy$$

$$\partial_{x_i} u = \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \frac{-2(x_i - y_i)}{4t} e^{-|x-y|^2/4t} g(y)\,dy$$

This is an absolutely convergent integral, so differentiating under integral sign was justified.

Higher x derivatives:

$$\partial_{x_1}^{a_1}\partial_{x_2}^{a_2}\cdots\partial_{x_n}^{a_n} u = \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \underbrace{\partial_{x_1}^{a_1}\cdots\partial_{x_n}^{a_n}\left(e^{-|x-y|^2/4t}\right)}_{\text{Polynomial} \times \text{Exponential in } y} g(y)\,dy$$

$\longrightarrow$ Convergent Integral

$\Rightarrow$ u is C^∞ in x

$$\partial_t^k u = \int_{\mathbb{R}^n} \partial_t^k \underbrace{\left[\frac{1}{(4\pi t)^{n/2}} e^{-|x-y|^2/4t}\right]}_{\text{Polynomial} \times \text{Exponential in } y} g(y)\,dy$$

$\Rightarrow$ All derivatives of u are well-defined

$\Rightarrow$ u is C^∞ in (x,t).

Thm If $U \subseteq \mathbb{R}^n$ is open and u solves $u_t - \Delta u$ in

$$U_T = \{(x,t) : x \in U, t \in (0,T)\},$$

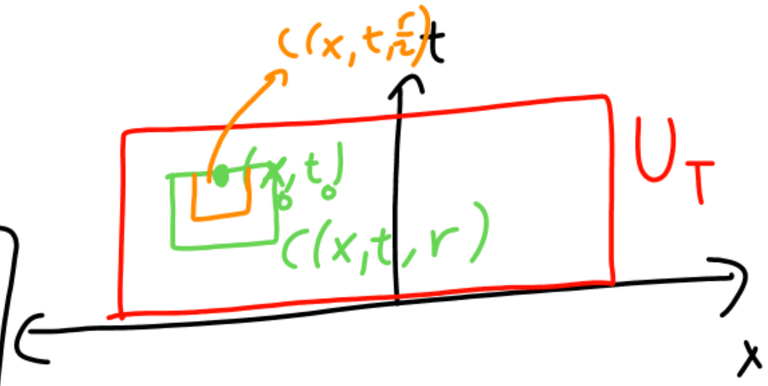
then $u \in C^\infty(U_T)$

Pf Don't have solution formula in this case.

To get around this, we localize.

Choose a cylinder

$$C(x_0, t_0, r) = \{(y,s) \in \mathbb{R}^n \times (0,T) : t_0 - r^2 < s \leq t_0, \ |y - x_0| < r\}$$



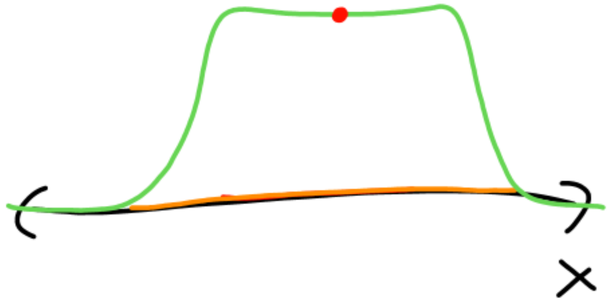
Let $\zeta(x,t)$ be a C^∞ function so that:

$$\zeta(x,t) = 1 \quad \text{in} \quad C(x_0,t_0,\tfrac{r}{2})$$

and $\zeta(x,t) = 0$ outside $C(x_0,t_0,r)$

and $0 \le \zeta \le 1$ everywhere

We call ζ a <u>cutoff function</u>.



Define $v(x,t) = u(x,t)\,\zeta(x,t)$

What PDE does v satisfy?

$$v_t = u_t\zeta + u\zeta_t, \qquad \nabla v = \nabla u\,\zeta + u\nabla\zeta$$

$$\Delta v = \Delta u\,\zeta + 2\nabla u\cdot\nabla\zeta + u\Delta\zeta$$

$$v_t - \Delta v = \underline{u_t \zeta} + u\zeta_t \underline{- \Delta u \zeta} - 2Du \cdot D\zeta - u\Delta\zeta$$

$$\zeta(u_t - \Delta u) = 0$$

$$= u(\zeta_t - \Delta\zeta) - 2Du \cdot D\zeta .$$

Let $f = u(\zeta_t - \Delta\zeta) - 2Du \cdot D\zeta \longrightarrow v_t - \Delta v = f$.

Note: $f = 0$ except in $C(x_0, t_0, r) \setminus C(x_0, t_0, \frac{r}{2})$.

Extend v to be defined on $\mathbb{R}^n \times [0, T]$ by setting $v = 0$ when $x \notin U$.

Solution formula:

$$u(x,t) = \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} e^{-|x-y|^2/4t} \underbrace{u(y,0)}_{=0 \text{ because of } \mathcal{S}} dy$$

$$+ \int_0^t \int_{\mathbb{R}^n} \frac{1}{(4\pi(t-s))^{n/2}} e^{-|x-y|^2/4(t-s)} f(y,s)\, dy\, ds$$

Differentiate u to any order:

$$\partial_{x_1}^{a_1} \cdots \partial_{x_n}^{a_n} u = \int_0^t \int_{\mathbb{R}^n} \underbrace{\frac{1}{(4\pi(t-s))^{n/2}} \left(\partial_{x_1}^{a_1} \cdots \partial_{x_n}^{a_n} e^{-|x-y|^2/4(t-s)} \right)}_{\text{Singularity at } x=y,\ t=s} f(y,s)\, ds$$

Integral doesn't see singularity since $f=0$ near $x=y$, $t=s$.

Time derivatives: similar argument.

Previous 2 theorems were <u>qualitative</u> regularity theorems.

<u>Thm</u> If $u_t - \Delta u = 0$ in U_T

and $C(x_0, t_0, r) \subset U_T$,

then

$$\max_{C(x_0, t_0, \frac{r}{2})} \left| \partial_{x_1}^{a_1} \cdots \partial_{x_n}^{a_n} \partial_t^k u \right| \leq C_{k,\ell} \left(\frac{1}{r^{\ell + 2k + n + 2}} + 1 \right) \int_{C(x_0, t_0, r)} |u| \, dx$$

where $\ell = a_1 + \cdots a_n$

Proof Sketch

First, assume $r = 1$, and

Let φ be a cutoff with

$$\varphi = 1 \quad \text{in } C(x_0, t_0, \tfrac{r}{2})$$

$$\varphi = 0 \quad \text{outside } C(x_0, t_0, \tfrac{3r}{4})$$

Let $v = u\varphi \longrightarrow v_t - \Delta v = f$ with $f = u(\varphi_t - \Delta\varphi) - 2Du \cdot D\varphi$

$$v(x,t) = \int_0^t \int_{\mathbb{R}^n} \frac{1}{(4\pi(t-s))^{n/2}} e^{-|x-y|^2/4(t-s)} f(y,s)\,dy\,ds$$

$$\partial_{x_i} v = \int_0^t \int_{\mathbb{R}^n} \frac{1}{(4\pi(t-s))^{n/2}} \frac{-2(x_i-y_i)}{4(t-s)} e^{-|x-y|^2/4(t-s)} \left[u(\zeta_t - \Delta\zeta) - 2\nabla u \cdot \nabla\zeta\right](y,s)\,dy\,ds$$

Int. by parts in term with ∇u:

$$\partial_{x_i} v = \int_0^t \int_{\mathbb{R}^n} \frac{-(x_i-y_i)}{2(4\pi(t-s))^{n/2}} \frac{1}{t-s} e^{-|x-y|^2/4(t-s)} \left[u(\zeta_t - \Delta\zeta) + 2u\Delta\zeta\right] dy\,ds$$

$$+ \int_0^t \int_{\mathbb{R}^n} \nabla_y \left[\frac{-(x_i-y_i)}{2(t-s)(4\pi(t-s))^{n/2}} e^{-|y-x|^2/4(t-s)}\right] u\nabla\zeta\,dy\,ds$$

Using that $D\xi$, $\Delta\xi$, ξ_t are bounded,

and equal to zero near singularity at

$(y,s)=(x,t)$,

one can show

$$|\partial_{x_i} v(x_0,t_0)| \leq C \int_{C(x_0,t_0,1)} |u|\,dy\,ds$$

Since $\xi_{x_i}=0$ at (x_0,t_0) $\Rightarrow$ $\partial_{x_i} v = u\xi_{x_i} + u_{x_i}\xi = u_{x_i}$ |

$$\Rightarrow |u_{x_i}(x_0,t_0)| \leq C \int_{C(x_0,t_0,1)} |u|\,dy\,ds$$

General r : let $w(x,t) = u(x_0 + rx, t_0 + r^2 t)$

$$\longrightarrow \quad w_t - \Delta w = r^2 u_t(x_0 + rx, t_0 + r^2 t) - r^2 \Delta u(x_0 + rx, t_0 + r^2 t)$$

$$= 0.$$

w defined in $C(0,0,1)$

if u defined in $C(x_0, t_0, r)$

Apply $r=1$ case to w:

$$|\partial_{x_i} w(x,t)| \leq C \int_{C(0,0,1)} |w| \, dx \, dt$$

$$\partial_{x_i} w = \partial_{x_i} u(x_0 + rx, t_0 + r^2 t)$$

$$= r (\partial_{x_i} u)(x_0 + rx, t_0 + r^2 t)$$

$$\int_{C(0,0,1)} |w|\,dx\,dt = \int_{C(0,0,1)} u(x_0+rx,\, t_0+r^2t)\,dx\,dt$$

$$y = x_0 + rx \longrightarrow dy = r^n\,dx$$

$$s = t_0 + r^2 t \longrightarrow ds = r^2\,dt$$

$$\int_{C(x_0,t_0,r)} u(y,s)\,\frac{dy\,ds}{r^{n+2}}$$

$$\Rightarrow \quad |u_{x_i}(x_0,t_0)| \le \frac{C}{r^{n+2+1}} \int_{C(x_0,t_0,r)} |u|\,dy\,ds$$

Higher derivatives: similar argument...

Q: Are solutions to $u_t - \Delta u = f$ smooth?

A: It depends on f.

Consider $f(t,x) = |x|$ $\quad n = 1$

Solution to $u_{xx} = f$

is not in C^3 since $|x|$, is not C^1.

If f is smooth, so is u.

Energy Methods: See HW and textbook.